

W-FDM: WAVELET-DOMAIN FEATURE DENOISING

Robust Object Detection Under Adverse Visual Conditions via Plug-In Frequency-Domain Feature Purification

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01 ABSTRACT & MOTIVATION

Object detectors trained on clean images suffer **significant accuracy degradation** when deployed in adverse real-world conditions like low-light environments, fog, and sensor noise.

We introduce **W-FDM** — a **lightweight, plug-in Wavelet Feature Denoising Module** inserted directly into the YOLOv8 neck (Layer 16). Unlike pixel-level preprocessing, W-FDM purifies deep feature representations in the frequency domain without requiring additional supervision signals.

- Domain Gap:** Standard detectors fail on ExDark-style low-light — mAP drops up to 30% vs. clean baseline on adverse test sets.
- Frequency Insight:** Degradation artifacts (blur, noise, haze) manifest disproportionately in high-frequency wavelet bands (*LH, HL, HH*), not in the semantic *LL* band.
- Targeted Fix:** By isolating and denoising only high-frequency feature subbands, we preserve object semantics while eliminating corruption artifacts.
- Plug-In Design:** Single module insertion — zero changes to backbone weights, no additional labels, end-to-end trainable.

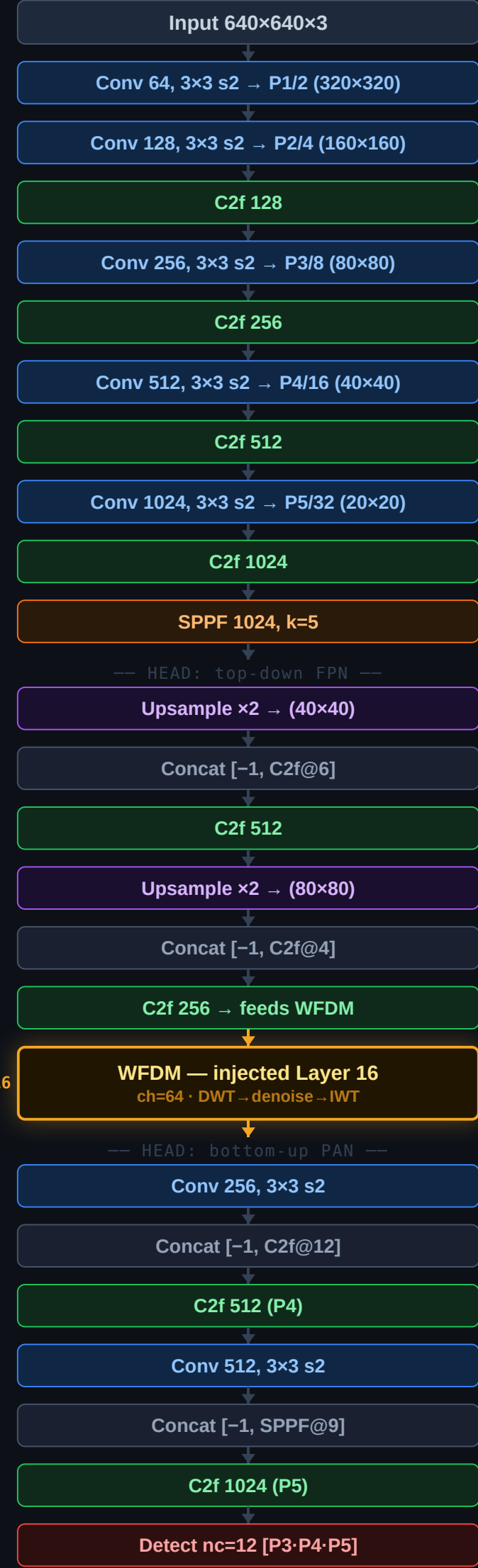
04 EXPERIMENTAL RESULTS

PRIMARY METRIC	SECONDARY METRIC		DATASET	
mAP@0.5 IoU threshold 0.5	mAP@0.5:0.95 COCO-style avg.		ExDark Low-light · 12 cls	
MODEL VARIANT	MAP@0.5	MAP@0.5:0.95	VS. BASELINE	NOTES
★ OURS YOLOv8n + W-FDM	0.404	0.234	+12.5% / +15.8%	Layer 16 inject + TTA
BASE YOLOv8n Baseline (fine-tuned)	0.359	0.202	- ref -	Trained on ExDark, no W-FDM
ZERO YOLOv8n Zero-Shot	0.0013	0.0003	-99.6%	Pretrained COCO, no fine-tune

Evaluation Protocol: All models evaluated on ExDark test split. W-FDM uses TTA (Test-Time Augmentation) at validation. Decision thresholds optimized via PR curve per class.

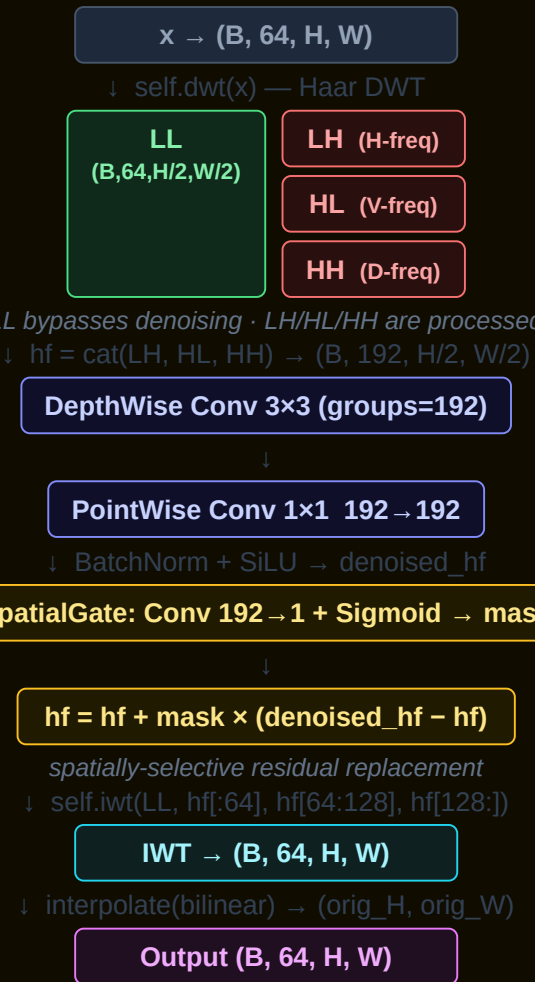
02 ARCHITECTURE: YOLOV8N + W-FDM INJECTION

FULL NETWORK (LAYER INDEX) · W-FDM MODULE — INTERNALS



Legend: Conv (blue), C2f (green), SPPF (orange), Upsample (purple), Concat (grey), Detect (red), W-FDM (yellow star)

W-FDM.FORWARD(X)



DWT — Haar Decomposition

$A = x[0::2, 0::2]$ $B = x[0::2, 1::2]$
 $C = x[1::2, 0::2]$ $D = x[1::2, 1::2]$
 $LL = (A+B+C+D) \times 0.5$
 $LH = (A+B-C-D) \times 0.5$
 $HL = (A-B+C-D) \times 0.5$
 $HH = (A-B-C+D) \times 0.5$

IWT — Haar Reconstruction

$out[0::2, 0::2] = (LL+LH+HL+HH) \times 0.5$
 $out[0::2, 1::2] = (LL+LH-HL-HH) \times 0.5$
 $out[1::2, 0::2] = (LL-LH+HL-HH) \times 0.5$
 $out[1::2, 1::2] = (LL-LH-HL+HH) \times 0.5$

DESIGN RATIONALE

- Why Layer 16 (after C2f@15)?**
Layer 15 is the first FPN merge of P3+P4 top-down features — the richest multi-scale representation. Inserting W-FDM here denoises before PAN bottom-up fusion can amplify corrupted features upward into the detect head.
- Why only HF bands (LH/HL/HH)?**
Fog, motion blur and sensor noise corrupt edge structure (high-freq) while the LL band preserves semantic object identity. Modifying LL would corrupt bounding box regression and classification features — so it bypasses the denoiser entirely.
- Spatial Gate Mask**
A single-channel sigmoid map allows the network to apply heavy denoising in corruption-dense regions while leaving clean areas untouched. This is critical for ExDark's heterogeneous corruption where some image regions are clean and others are severely degraded.
- Depthwise Separable Convolution**
The DW+PW conv block on the concatenated HF subbands (192ch) is parameter-efficient — each subband is processed independently before cross-channel mixing, reducing compute while maintaining denoising capacity.
- Residual Replacement:**
 $hf = hf + mask \times (denoised_hf - hf)$
The gated residual form ensures that if the spatial mask is near-zero, the original features pass through unchanged — making W-FDM a strict no-op in clean conditions.

03 TRAINING & OPTIMIZATION

End-to-end training on **ExDark** — 7,363 real low-light images across 12 object classes. Phase 2 fine-tuning optimized specifically for the spatial gate convergence.

OPTIMIZER AdamW — Stable gate convergence	WEIGHT DECAY 0.01
LEARNING RATE 0.0005	
AUGMENTATIONS Mosaic + Mixup + hsv_v=0.4	
VALIDATION STRATEGY TTA (Test-Time Augmentation) for mAP stability	
DATASET ExDark (real low-light)	CLASSES 12 object types
W-FDM CHANNELS (CH) 64 — Inserted at FPN output (80×80 spatial scale)	

05 CONCLUSIONS & FUTURE WORK

- ✓ **Frequency-domain feature purification works.** Denoising only high-frequency wavelet subbands (LH, HL, HH) while bypassing LL preserves semantic content and improves detection under adverse conditions.
- ✓ **Spatial gating is critical.** The adaptive sigmoid mask selectively suppresses noise in corrupted regions while leaving clean areas untouched — enabling robust generalization across ExDark's heterogeneous scene types.
- ✓ **Layer 16 is the optimal injection point.** Inserting after the first FPN merge (P3+P4) captures the richest multi-scale features before PAN propagation amplifies corruption.
- ✓ **Plug-in modularity.** W-FDM requires no changes to backbone weights, no additional supervision, and adds minimal parameters — compatible with any YOLO-family architecture.
- 🔭 **Future:** Multi-scale injection (P4, P5), learnable wavelet bases, fog/rain/snow domain generalization, and real-time deployment profiling on edge hardware.

CORE INSIGHT

Adverse visual conditions corrupt *high-frequency feature structure*, not semantic identity. Targeted frequency-domain denoising — guided by a learned spatial attention mask — is a computationally efficient and architecturally clean solution to bridging the clean-to-degraded domain gap.

06 QUALITATIVE RESULTS — W-FDM DETECTION EXAMPLES ON EXDARK

